

Last Name:	First Name:	ID:
------------	-------------	-----

Question	Maximum mark	Actual mark
1	8	
2	8	
3	8	
4	5	
5	6	
6	5	
7	5	
8	10	
9	5	
Total	60 pts	60 pts

Question 1.

8 pts

Assume an experiment where a ball is drawn from a bag containing a red (R), blue (B), and green (G) ball, followed by spinning a spinner with outcomes labeled 1 through 4.

- (a) (2 pts) List all possible simple events (elements) of the resulting sample space (use R, B, G for ball colors and 1 to 4 for spinner outcomes).
- (b) (3 pts) Write down the simple events (elements) that constitute the following events:
- (1.5 pts) $C = \{\text{a red or blue ball and an even number on the spinner}\}$,
 - (1.5 pts) $D = \{\text{a green ball and a number greater than 2}\}$.
- (c) (3 pts) Write down the sub-spaces that represent the following events: $C \cap D$, $C \cup D$, and $C^c \cap D^c$.

Solution 1.

The experiment consists of drawing a ball from a bag (with colors R = red, B = blue, G = green) and spinning a spinner with outcomes $\{1, 2, 3, 4\}$.

(a) Sample Space:

$$S = \{(R, 1), (R, 2), (R, 3), (R, 4), (B, 1), (B, 2), (B, 3), (B, 4), (G, 1), (G, 2), (G, 3), (G, 4)\}$$

(b) Events:

- $C = \{\text{a red or blue ball and an even number on the spinner}\}$
Red or blue: R, B; Even numbers: 2, 4

$$C = \{(R, 2), (R, 4), (B, 2), (B, 4)\}$$

- $D = \{\text{a green ball and a number greater than 2}\}$
Green ball: G; Number > 2 : 3, 4

$$D = \{(G, 3), (G, 4)\}$$

(c) Event Operations:

- $C \cap D$: Events common to both C and D

$$C \cap D = \emptyset$$

- $C \cup D$: All events in either C or D

$$C \cup D = \{(R, 2), (R, 4), (B, 2), (B, 4), (G, 3), (G, 4)\}$$

- $C^c \cap D^c$: Events not in C and not in D

$$C^c \cap D^c = S \setminus (C \cup D)$$

$$C^c \cap D^c = \{(R, 1), (R, 3), (B, 1), (B, 3), (G, 1), (G, 2)\}$$

Question 2.

8 pts

1. (2 pts) A school is selecting students for a science quiz. How many ways can the teacher choose 4 students from a group of 12 candidates?
2. There are 7 chocolate muffins and 5 vanilla muffins in a bakery display. A customer picks 6 muffins at random. Find the probability that
 - i. (3 pts) there will be exactly 4 chocolate muffins among the chosen muffins;
 - ii. (3 pts) there will be less than 3 chocolate muffins among the chosen muffins?

Solution 2.

1. Since we do not distinguish the students, this is a combination problem. We have

$$\binom{12}{4} = \frac{12!}{4!8!} = 495.$$

2. for these questions,

- i. One has

$$p = \frac{\binom{7}{4} \binom{5}{6-4}}{\binom{12}{6}} = \frac{35 \cdot 10}{924} = \frac{350}{924} \approx 0.3787.$$

- ii. Less than 3 chocolate muffins means 1 or 2 chocolate muffins. Hence, the requested probability is

$$p = \frac{\binom{7}{1} \binom{5}{5} + \binom{7}{2} \binom{5}{4}}{\binom{12}{6}} = \frac{7 + 105}{924} = \frac{112}{924} \approx 0.1212.$$

Question 3.

8 pts

1. (2 pts) What is meant by saying events A and B are independent ?
2. (6 pts) Suppose $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$.
 - i. (2 pts) Find $P(A \cup B)$
 - ii. (2 pts) Are A and B mutually exclusive? Justify?
 - iii. (2 pts) Are A and B independent? Justify?

Solution 3.

1. A and B are independent if and only if $P(A \cap B) = P(A) \cdot P(B)$.
2. Given that $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$
 - i. To find $P(A \cup B)$, we use the formula:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substituting the values:

$$P(A \cup B) = 0.4 + 0.3 - 0.12 = 0.58$$

- ii. Events A and B are mutually exclusive if $P(A \cap B) = 0$. Since $P(A \cap B) = 0.12$ (which is greater than 0).
Hence, events A and B are **not** mutually exclusive.

iii. Events A and B are independent if:

$$P(A \cap B) = P(A) \times P(B)$$

Calculating $P(A) \times P(B)$:

$$P(A) \times P(B) = 0.4 \times 0.3 = 0.12$$

Since $P(A \cap B) = 0.12$ and $P(A) \times P(B) = 0.12$, we find that:

$$P(A \cap B) = P(A) \times P(B)$$

Thus, events A and B are independent.

Question 4.

5 pts

A factory has a batch of 24 juice bottles, of which 6 are spoiled. If 4 bottles are selected at random and removed one after another **without replacement**, what is the probability that all 4 selected bottles are spoiled?

Solution 4.

Let

- A be the event that the first bottle is spoiled,
- B be the event that the second bottle is spoiled,
- C be the event that the third bottle is spoiled,
- and D be the event that the forth bottle is spoiled,
- then

$$\mathbb{P}(A) = \frac{6}{24}, \mathbb{P}(B | A) = \frac{5}{23}, \mathbb{P}(C | A \cap B) = \frac{4}{22}, \quad P(D | B \cap C \cap D) = \frac{3}{21},$$

and using multiplication rule, we get

$$\begin{aligned} \mathbb{P}(A \cap B \cap C) &= \mathbb{P}(A) \cdot \mathbb{P}(B | A) \cdot \mathbb{P}(C | A \cap B) \cdot P(D | B \cap C \cap D) \\ &= \frac{6}{24} \cdot \frac{5}{23} \cdot \frac{4}{22} \cdot \frac{3}{21} = \frac{5}{3542} \approx 0.00141. \end{aligned}$$

Question 5.

6 pts

On rainy days, Joe is late to work with probability 0.3; on non-rainy days, he is late with probability 0.1. With probability 0.7, it will rain tomorrow.

1. (3 pts) Find the probability that Joe will not be late tomorrow.
2. (3 pts) Given that Joe was not late, what is the conditional probability that it rained?

Solution 5.

Let A denote Joe is late, and let B denote the weather is rainy. We have

$$P(A|B) = 0.3, \quad P(A|B^c) = 0.1, \quad P(B) = 0.7.$$

1. $P(A) = P(B) \cdot P(A|B) + P(B^c) \cdot P(A|B^c) = 0.7 \cdot 0.3 + 0.3 \cdot 0.1 = 0.24$. Therefore,

$$P(A^c) = 1 - P(A) = 0.76.$$

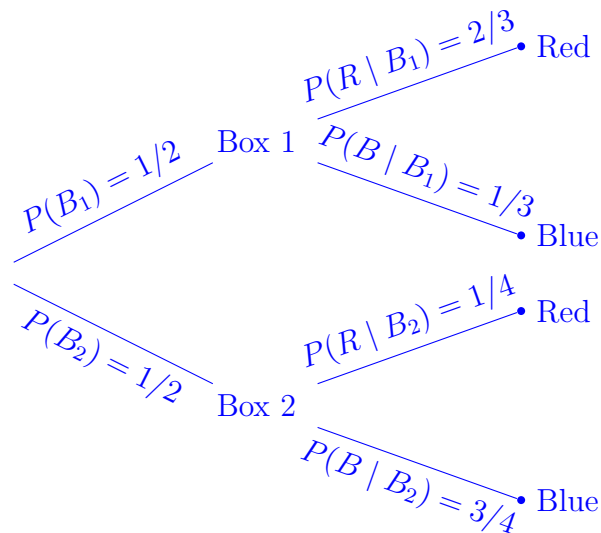
2. By definition,

$$P(B|A^c) = \frac{P(A^c|B)P(B)}{P(A^c)} = \frac{(1 - 0.3) \cdot 0.7}{0.76} = \frac{49}{76} = 0.644.$$

Question 6.

5 pts

Box 1 contains 2 red balls and 1 blue ball. Box 2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, box 1 is selected and a ball is drawn. If it falls tails up, box 2 is selected and a ball is drawn. Find the probability of selecting a red ball.

Solution 6.

Based on the tree diagram the desired probability is given by

$$P(R) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{4} = \frac{2}{6} + \frac{1}{8} = \frac{11}{24} \approx 0.4584.$$

Question 7.

5 pts

Let X be a discrete random variable with the following probability distribution,

$$f(x) = \frac{e^{-3}3^x}{x!}, \quad x = 0, 1, 2, \dots$$

(a) (3 pts) Verify that $f(x)$ is a valid probability mass function.

(b) (2 pts) Find the conditional probability $P(X < 5 \mid X \geq 3)$.

Solution 7.

(a) To verify $f(x)$ is a valid pmf, we check:

$$\sum_{x=0}^{\infty} f(x) = \sum_{x=0}^{\infty} \frac{e^{-3} 3^x}{x!} = e^{-3} \sum_{x=0}^{\infty} \frac{3^x}{x!} = e^{-3} e^3 = 1$$

Thus, $f(x)$ is a valid probability mass function.

(b) Find the conditional probability

$$P(X < 5 \mid X \geq 3) = \frac{P(3 \leq X < 5)}{P(X \geq 3)} = \frac{P(X = 3) + P(X = 4)}{P(X \geq 3)}$$

Calculate numerator terms:

$$P(X = 3) = \frac{e^{-3} 3^3}{3!} = e^{-3} \frac{27}{6} = 0.0498 \times 4.5 = 0.2241$$

$$P(X = 4) = \frac{e^{-3} 3^4}{4!} = e^{-3} \frac{81}{24} = 0.0498 \times 3.375 = 0.1681$$

Sum numerator:

$$P(3 \leq X < 5) = 0.2241 + 0.1681 = 0.3922$$

Calculate denominator:

$$P(X \leq 2) = e^{-3} \left(1 + 3 + \frac{9}{2} \right) = 0.0498 \times 8.5 = 0.4233$$

Thus,

$$P(X \geq 3) = 1 - 0.4233 = 0.5767$$

Finally,

$$P(X < 5 \mid X \geq 3) = \frac{0.3922}{0.5767} = 0.6802$$

Question 8.

A discrete random variable X has the probability distribution :

10 pts

$$g(x) = \frac{|x - 2|}{k}, \text{ for } x = -1, 0, 1, 3.$$

- (a) (3 pts) Find the appropriate k
- (b) (5 pts) Find the cumulative distribution function F , and plot its graph
- (c) (2 pts) Compute $P(X < 1)$

Solution 8.

(a) We need to check the following two conditions:

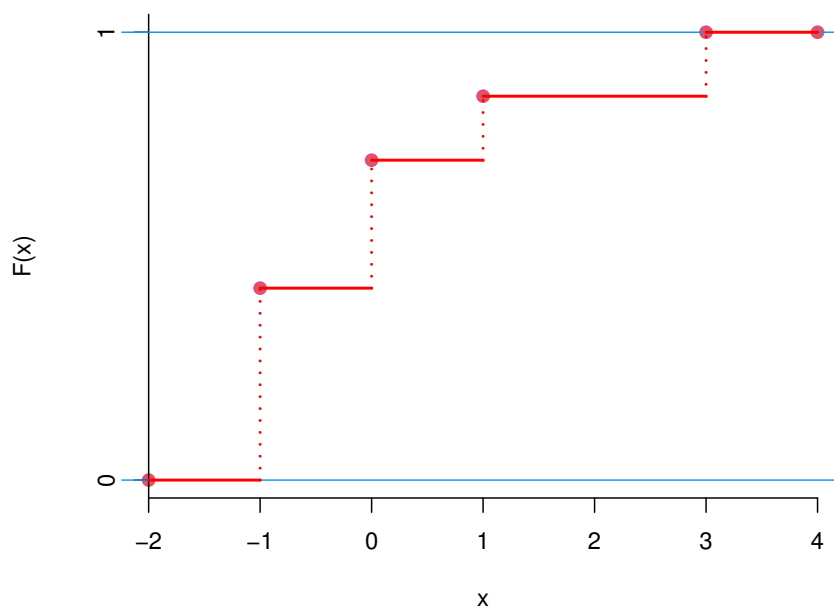
- $g(x) \geq 0 \implies k > 0.$
-

$$\begin{aligned} \sum_{x=-1,0,1,3} g(x) = 1 &\iff \frac{|-1-2|}{k} + \frac{|0-2|}{k} + \frac{|1-2|}{k} + \frac{|3-2|}{k} = 1 \\ &\iff \frac{1}{k} (3 + 2 + 1 + 1) = 1 \\ &\iff \boxed{k = 7} \end{aligned}$$

(b) The cdf is given by

$$F(x) = \begin{cases} 0 & \text{for } x < -1 \\ \frac{3}{7} & \text{for } -1 \leq x < 0 \\ \frac{5}{7} & \text{for } 0 \leq x < 1 \\ \frac{6}{7} & \text{for } 1 \leq x < 3 \\ 1 & \text{for } x \geq 3 \end{cases}$$

Cdf for exercise 10.



(c) The requested probability is

$$\begin{aligned} \mathbb{P}(X < 1) &= \mathbb{P}(X = -1) + \mathbb{P}(X = 0) \\ &= \frac{|-1-2|}{7} + \frac{|0-2|}{7} \\ &= \frac{3}{7} + \frac{2}{7} = \frac{5}{7} \end{aligned}$$

Question 9.

5 pts

The cumulative distribution function (CDF) of a discrete random variable X is defined as follows:

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2 & 0 \leq x < 1 \\ 0.5 & 1 \leq x < 2 \\ 0.7 & 2 \leq x < 3 \\ 0.9 & 3 \leq x < 4 \\ 1.0 & x \geq 4 \end{cases}$$

- (a) (4 pts) Determine the probability mass function (PMF), $P(X = x)$, for each value.
- (b) (1 pts) Verify that the PMF values sum to 1.

Solution 9.

(a) To find the PMF, we use:

$$P(X = x) = F(x_i) - F(x_{i-1})$$

We compute:

$$\begin{aligned} P(X = 0) &= F(0) - F(x < 0) = 0.2 - 0 = 0.2 \\ P(X = 1) &= F(1) - F(x < 1) = 0.5 - 0.2 = 0.3 \\ P(X = 2) &= F(2) - F(x < 2) = 0.7 - 0.5 = 0.2 \\ P(X = 3) &= F(3) - F(x < 3) = 0.9 - 0.7 = 0.2 \\ P(X = 4) &= F(4) - F(x < 4) = 1.0 - 0.9 = 0.1 \end{aligned}$$

So the PMF is:

x	0	1	2	3	4
$P(X = x)$	0.2	0.3	0.2	0.2	0.1

(b) Add all PMF values to verify they sum to 1:

$$0.2 + 0.3 + 0.2 + 0.2 + 0.1 = 1.0 \quad \checkmark$$